

Rectangular-Billiard Branch Coefficients and Pairwise-Transverse Aspect Crossings

Route-A structural certificate C167

Abstract

Rectangular Dirichlet billiards provide a natural one-parameter deformation of the square half-wave trace, but their shell multiplicities change when lattice directions collide. We prove a complete source-side statement for every aspect $\alpha > 0$. If $R_\alpha(E) = \#\{(m, n) \in \mathbb{Z}^2 : m^2 + \alpha^2 n^2 = E\}$, then the full Abel trace at $t = 2\sqrt{E}$ has the canonical boundary coefficient $\alpha e^{i\pi/4} R_\alpha(E)/(8\pi E^{1/4})$ under $\epsilon^{3/2}$ normalization; negative time gives its conjugate. The proof uses the exact rectangular Poisson formula, a finite shell split, and an ϵ -uniform summable tail; the simultaneous two-boundary pole at $\beta = 4$, $E = 4$ remains lower order. Writing $\beta = \alpha^2$, distinct absolute lattice directions collide exactly at positive rational values $\beta = (m'^2 - m^2)/(n^2 - n'^2)$; every pairwise collision is transverse, and irrational β therefore has sign-only degeneracy. For $\beta = u/v$, every shell is a fibre $vm^2 + un^2 = N$. Exact ledgers through coordinate 24 and four convergence rows are regression sentinels rather than proofs. The coefficient aggregates clean families and is not an isolated stability amplitude or a target divisor law. No arithmetic Euler, root-number, automorphy, Hilbert–Pólya, or Route-B claim is made.

Keywords: Abel half-wave trace; clean-family multiplicity; Poisson duality; rational collision stratum; boundary normalization; spectral geometry.

中文摘要：矩形狄利克雷台球把正方形半波迹扩展为一个连续形变族，其格点壳层重数会随纵横比改变。本文对任意 $\alpha > 0$ 证明：在壳层时间 $2\sqrt{E}$ ，完整阿贝尔迹经 $\epsilon^{3/2}$ 归一化后收敛到 $\alpha e^{i\pi/4} R_\alpha(E)/(8\pi E^{1/4})$ ，负时间给出复共轭。证明从矩形泊松公式出发，分离匹配壳层，并以二维可求和尾项控制其余项。令 $\beta = \alpha^2$ ，两个不同绝对格点方向只会在正有理 β 处碰撞；无理 β 仅保留坐标符号简并。当 $\beta = u/v$ 时，所有壳层由 $vm^2 + un^2 = N$ 的纤维精确描述。进一步，碰撞处能量差与时间差的一阶导数均非零，故每个成对碰撞都是横截的； $\beta = 4$, $E = 4$ 的双轴与双边界重合也不改变极限系数。有限格点表与数值极限只作回归检查。所得系数计数光滑轨道族，既不是孤立稳定性振幅，也不导出目标算术结构。

关键词：阿贝尔半波迹；光滑族重数；泊松对偶；有理碰撞层；边界归一化；谱几何。

1 Source family

For $\Re s > 0$, separation of variables gives

$$W_\alpha(s) = \sum_{j,k \geq 1} e^{-\pi s \sqrt{j^2 + k^2/\alpha^2}}.$$

Two-dimensional Poisson summation yields

$$W_\alpha(s) = \frac{\alpha s}{2\pi} \sum_{m,n \in \mathbb{Z}} (s^2 + 4(m^2 + \alpha^2 n^2))^{-3/2} - \frac{1}{4} - \frac{1}{2(e^{\pi s} - 1)} - \frac{1}{2(e^{\pi s/\alpha} - 1)}. \quad (1)$$

The dual series is locally normally convergent in the right half-plane.

2 Canonical shell coefficient

Let $E > 0$ be represented by the dual lattice and put

$$R_\alpha(E) = \#\{(m, n) \in \mathbb{Z}^2 : m^2 + \alpha^2 n^2 = E\}.$$

At $t_E = 2\sqrt{E}$ the complete trace satisfies

$$\lim_{\epsilon \downarrow 0} \epsilon^{3/2} W_\alpha(\epsilon - it_E) = \frac{\alpha e^{i\pi/4} R_\alpha(E)}{8\pi E^{1/4}}. \quad (2)$$

Negative time gives the complex conjugate. For a matching vector, $s^2 + 4E = \epsilon(\epsilon - 2it_E)$; the principal power in (1) gives one copy of the coefficient in (2). Fixing α and t_E , all sufficiently large nonmatching terms are bounded by a constant times $(m^2 + n^2)^{-3/2}$. This is summable on $\mathbb{Z}^2$, while the finite remainder stays bounded. Multiplication by $\epsilon^{3/2}$ kills both. The two boundary subtractions have at most simple poles and therefore also vanish after normalization. This remains true when both subtraction clocks coincide: they form a sum of simple poles rather than a second-order pole. At the hostile control $\beta = 4$, $E = 4$, the dual shell contains $(\pm 2, 0)$ and $(0, \pm 1)$ while both boundary terms are singular. Their normalized contribution is still $O(\epsilon^{1/2})$, so (2) retains the full multiplicity four.

3 Aspect collision law

Write $\beta = \alpha^2$. Distinct absolute representatives (m, n) and (m', n') collide only if $n^2 \neq n'^2$, in which case

$$\boxed{\beta = \frac{m'^2 - m^2}{n^2 - n'^2} > 0.} \quad (3)$$

Thus every non-sign collision parameter is positive rational, and irrational β has sign-only multiplicity. If $\beta = u/v$ in lowest terms, the complete shells are precisely the fibres $vm^2 + un^2 = N$.

At a collision β_0 , the squared-energy and time separations obey

$$\frac{d}{d\beta}(E_p - E_{p'}) = n^2 - n'^2 \neq 0, \quad \left. \frac{d}{d\beta}(t_p - t_{p'}) \right|_{\beta_0} = \frac{n^2 - n'^2}{\sqrt{E_0}} \neq 0. \quad (4)$$

Multiple fibres have distinct n^2 slopes for distinct absolute representatives. Hence their pairwise crossings are transverse. Side interchange also gives the exact reciprocal-aspect identity

$$W_\alpha(s) = W_{1/\alpha}(s/\alpha), \quad (5)$$

which transports the same source family with the corresponding clock rescaling.

Exact coordinate-24 ledgers and high-precision limits are sentinels only. Equation (3), not that ledger, proves the irrational no-collision statement; it does not imply a uniform irrational shell gap. Likewise $vm^2 + un^2 = N$ is an exact fibre classification, not a claimed universal divisor formula. The strict tuple is

$$(A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION). \quad (6)$$

The coefficient aggregates clean source families. We claim no isolated stability determinant, target trace/divisor/counting law, arithmetic local or Euler factor, root number, automorphy, Hilbert–Pólya construction, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

Data availability. All claim-bearing code and data are included; no external dataset is used. **Ethics.** No human participants, animals, or sensitive personal data are involved. **Author contributions/CRedit.** Anonymous technical certificate; Conceptualization, Formal analysis, Software, Validation, and Writing are recorded at package level. **Competing interests.** None known. **Funding.** No external funding is reported. **AI-use disclosure.** An AI coding assistant supported proof planning, drafting, and code review. Packaged exact checks validate quantitative claims; the assistant was not treated as an external peer reviewer.